

Notes: Wollmann (AER, 2018)

Trucks without Bailouts: Equilibrium Product Characteristics for Commercial Vehicles

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1 Big picture

- **Question.** How much does allowing *product-level* entry and exit change predicted post-merger or post-bankruptcy outcomes in a differentiated product industry?
- **Setting.** The paper studies the U.S. commercial vehicle market. The key institutional fact is that *firm entry and exit are rare*, but *model-level entry and exit are common*. That makes product offerings, not firm counts, the main adjustment margin.
- **Core challenge.** Standard merger or bailout counterfactuals often hold product offerings fixed and let only prices adjust. In this market, that is likely wrong because surviving firms can introduce new truck models or retire overlapping ones very quickly.
- **Main theoretical idea.** Firms play a two-stage game. First, they choose which products to offer, facing sunk entry costs and scrap values from retirement. Second, conditional on the chosen product menu, they set prices in a differentiated-demand oligopoly.
- **Main empirical strategy.** Estimate demand and marginal costs in stage 2, recover expected profits from any candidate product menu, and then use moment inequalities plus exogenous demand shifters to bound stage-1 sunk costs of adding and dropping products.
- **Identification of sunk costs.** The paper uses variation in the composition of potential buyers—especially construction demand and freight-related cab-length regulation—to generate plausibly exogenous shifts in the profitability of specific truck types.
- **Main quantitative result.** Ignoring product entry and exit can hugely overstate markup increases and output losses. Under a Ford acquisition of GM and Chrysler, the markup of the most affected model rises by **62.1%** if products are held fixed, but only **16.8%** once product entry and exit are allowed. Under liquidation, market output falls **11.2%** if products are fixed, but only **1.6%** when product offerings adjust.
- **Policy result.** Endogenous product repositioning can be *pivotal* for the bailout decision. In the liquidation counterfactual, compensating variation is **\$253 million** when entry/exit is ignored but only **\$28 million** when it is allowed.
- **Economic intuition.** Policies such as liquidation or acquisition create local pockets of market power. But these same pockets create profit opportunities exactly where surviving rivals want to enter new models. Entry increases substitution possibilities and disciplines prices.
- **Takeaway.** In industries with stable firm identities but flexible product menus, the right policy model must let products move. Otherwise, merger, bankruptcy, and bailout analysis can be badly misleading.

2 Data and measurement (key objects)

2.1 Market definition and economic environment

- The paper studies the **commercial vehicle** segment of the U.S. auto industry: on-road vehicles with gross vehicle weight rating (GWR) above **10,000 lbs.**
- This segment is economically important: about **10%** of total U.S. automotive sales, while the auto industry itself is about **4%** of GDP.

- The market is ideal for the paper’s question because:
 - there has been essentially **no firm entry or exit** for decades,
 - ownership is concentrated among roughly ten firms,
 - but product offerings change frequently,
 - and the body-on-frame production technology makes model changes relatively fast and modular.

2.2 Main data sources

The paper combines several datasets.

- **Product characteristics and prices:** annual issues of *The Truck Blue Book*, 1986–2012. These data contain model names, brands, and detailed characteristics, including list prices.
- **Unit sales:** R. L. Polk new vehicle registration data, merged to the product file.
- **Buyer microdata:** TIUS/VIUS Census surveys, which identify who buys what type of truck, including buyer industry and location.
- **Market-size and buyer-composition shifters:** County Business Patterns and U.S. Department of Transportation road-mileage data.
- **Costs:** Bureau of Labor Statistics wage data matched to product assembly locations.

Important measurement issue. The paper observes *MSRP*, not transaction prices. This is a limitation, but it is common in the literature, and the author checks whether implied margins are at least consistent with reported operating margins.

2.3 Key product characteristics

The paper reduces commercial vehicles to a short list of characteristics that matter for buyer demand and firm supply.

- **GWR (gross vehicle weight rating).** This is the central characteristic. It determines load capacity and is the main way both regulators and industry participants classify trucks.
- **Cab type.** There are three main cab types:
 - **Conventional:** long hood, smoother and more comfortable, but less maneuverable.
 - **Cab-over-engine (cabover):** flat front, highly maneuverable and visible, but less comfortable.
 - **Compact-front-end:** compromise between the two.
- **Long/extended cab option.** Mostly relevant for long-haul uses.

Why these characteristics are useful. They generate clear and intuitive links between buyer type and product choice, which is exactly what the demand system needs.

2.4 Buyer heterogeneity

A major strength of the paper is that it does not treat truck buyers as homogeneous.

- Buyers from **freight industries** prefer high-GWR vehicles.
- Buyers from **construction industries** prefer medium GWR ranges.
- **Urban buyers** value cabovers more because visibility and turning radius matter more in dense areas.

Figure 1 in the paper makes these patterns visual:

- Panel A shows expected GWR differs a lot by buyer industry.
- Panel B shows cabovers are unpopular overall, but heavily used in dense urban areas.

2.5 Descriptive facts from Table 1

Table 1 shows several facts that motivate the whole project.

- The total number of product offerings varies substantially over time: from about **62** to **85**, with averages around **75–78**.
- Average prices are roughly **\$70,000–\$76,000** (in 2005 dollars), depending on subperiod.
- Annual unit sales vary a lot, from about **165,678** to **584,057**.
- Buyer composition also moves: freight-related industries account for roughly **22–32%** of potential buyers, construction-related industries for **36–50%**, and business/personal services for **23–37%**.

Interpretation. The market is large, cyclical, and compositionally shifting. Those shifts are exactly what can make some truck types more profitable to add and others more attractive to retire.

2.6 What is observed and what is not

- **Observed:** prices, product characteristics, quantities, buyer composition, wages, and firm/product identities.
- **Not directly observed:** match-specific taste shocks, marginal-cost shocks, sunk-cost disturbances, and the dynamic continuation values that would appear in a full dynamic programming problem.

This distinction is important because the paper’s method is built precisely to recover these latent objects from equilibrium restrictions.

3 Models and empirical specifications (all equations + interpretation)

3.1 Stage 2 demand: buyer utility

Each buyer r chooses among truck products j and an outside option. Utility from product j is

$$U_{rj} = x_j(\beta_x + \beta_{xo}z_{ro} + \beta_{xu}z_{ru}) + p_j\beta_p + \xi_j + \varepsilon_{rj}. \quad (1)$$

Here:

- x_j is the vector of truck characteristics excluding price,
- z_{ro} and z_{ru} are observed and unobserved buyer attributes,
- p_j is price,
- ξ_j is a product-specific taste shock,
- ε_{rj} is an idiosyncratic logit error.

Interpretation.

- Trucks are differentiated by characteristics, not just by brand.
- Buyer type matters: for example, freight firms care more about GWR, and urban buyers care more about cabover design.
- The logit shock gives closed-form choice probabilities, which makes the large-scale pricing and counterfactual exercises feasible.

After integrating over simulated buyers, the model produces market shares s_{jt} , and quantities are

$$q_{jt} = M_t s_{jt}, \quad (2)$$

where M_t is market size.

3.2 Stage 2 pricing: firm profits and marginal costs

Given a product menu, each firm f chooses prices to maximize

$$\Pi_{ft} = \sum_{j \in J_{ft}} (p_{jt} - mc_{jt}) s_j(x_t, p_t, z_t; \beta, \xi_t) M_t. \quad (3)$$

Marginal costs are modeled log-linearly:

$$\log(mc_{jt}) = [x_{jt}, w_t]\gamma + \omega_{jt}, \quad (4)$$

where x_{jt} are product characteristics, w_t includes wages and time effects, and ω_{jt} is an unobserved cost shock.

Interpretation.

- Stage 2 is a standard differentiated-product Bertrand pricing problem.
- Once demand is estimated, one can back out marginal costs from the pricing first-order conditions.
- This stage tells the econometrician how profitable any given product menu would be.

3.3 Stage 1 product offering problem

Before prices are set, firms choose which products to offer. The key object is the expected second-stage profit from any candidate menu:

$$\pi(J_{ft}, J_{-f,t}, z_t, w_t) = \int \Pi(J_{ft}, J_{-f,t}, z_t, w_t; \beta, \gamma, \xi', \omega') dF_{\xi'} dF_{\omega'}. \quad (5)$$

Sunk costs for product j are modeled as

$$SC_{fjt, J_{f,t-1}} = x'_j \tilde{\theta}_{f, x_j, t} [\mathbf{1}\{j \in J_{ft}, j \notin J_{f,t-1}\} + \lambda \mathbf{1}\{j \notin J_{ft}, j \in J_{f,t-1}\}]. \quad (6)$$

Interpretation.

- If a product is newly introduced, the firm pays an entry cost.
- If a product is retired, the firm gets back a fraction of that cost as a scrap value.
- The scrap-value parameter λ scales retirement payoffs relative to entry costs.

3.4 Why the paper does not solve the full dynamic program

A full dynamic product-positioning model with many potential product types would require tracking an enormous state space. The paper argues this is not how managers actually plan in practice. Instead, the paper follows the idea that managers often use **hurdle rates** and approximate capital budgeting rules.

This allows the author to rewrite the product choice problem in a way that compares:

- the expected profit consequences of adding or dropping a product, and
- a *scaled* sunk-cost term.

The important implication is that the empirical exercise estimates these scaled sunk costs directly, and those are enough for policy counterfactuals.

3.5 Stage 1 decision rules: keep, enter, or exit

If a firm currently offers product j , then keeping it must beat dropping it:

$$\Delta\pi(J_{ft}, J_{ft} \setminus j, J_{-f,t}, z_t, w_t) + x'_j(\text{stay/exit term}) \mathbb{E}[\theta_{f, x_j, t} \mid \mathcal{I}_{ft}] \geq 0. \quad (7)$$

If a firm does *not* currently offer product j , then staying out must beat entering it:

$$\Delta\pi(J_{ft}, J_{ft} \cup j, J_{-f,t}, z_t, w_t) - x'_j(\text{entry/continuation term}) \mathbb{E}[\theta_{f, x_j, t} \mid \mathcal{I}_{ft}] \geq 0. \quad (8)$$

Interpretation.

- Observed product menus must satisfy one-product deviation inequalities.
- This is the paper's key computational simplification: it uses *necessary conditions* for Nash equilibrium rather than trying to enumerate every possible equilibrium in a gigantic positioning game.

3.6 Why moment inequalities are natural here

The stage-1 game almost surely has multiple equilibria, and the feasible product space is huge. Because of that, the paper uses moment inequalities rather than exact likelihood-based estimation. The generic sample inequalities look like

$$\mathbb{E}\left[h_{f,x_j,t} \mathbf{1}\{j \in J_{ft}\} (\Delta \hat{\pi}_{drop} + \text{sunk-cost term})\right] \geq 0, \quad (9)$$

and

$$\mathbb{E}\left[h_{f,x_j,t} \mathbf{1}\{j \notin J_{ft}\} (\Delta \hat{\pi}_{add} - \text{sunk-cost term})\right] \geq 0. \quad (10)$$

The weighting functions $h_{f,x_j,t}$ select subsamples in which entry or exit for a particular product type is particularly likely.

Interpretation.

- The paper is not asking the data to identify one exact sunk-cost number from one exact equilibrium.
- It only asks that the observed menus be rationalizable by one-product deviations in carefully chosen states of the world.
- This is exactly why the demand shifters in Section IV matter so much.

3.7 Stage 2 estimation in practice

The demand system is estimated using GMM, closely following Petrin-style logic.

- **Cost shifter instrument:** wages by production location.
- **Markup shifter instrument:** local competitive environment and product overlap.
- **Micro moments:** moments matching buyer attributes to purchased product types, such as industry-by-GWR and urban-by-cabover patterns.

Once demand is estimated, the pricing first-order conditions imply marginal costs, and the paper then regresses the recovered $\log(mc)$ on observables to estimate γ .

3.8 Counterfactual policy experiments

The main policy comparison is between the actual **bailout** and three alternatives:

- acquisition of GM and Chrysler by **Ford**,
- acquisition by **PACCAR**,
- **liquidation**.

For each policy, the paper compares two economic models:

1. a model where only prices adjust,
2. a model where both prices and product offerings adjust.

To compute product-offering equilibria, the paper uses a best-response learning process similar to Lee and Pakes (2009): firms update product choices sequentially until no firm wants to deviate. Outcomes are averaged over 30 draws of sunk-cost shocks.

Interpretation.

- This is not a perfect global equilibrium selection procedure.
- But it is computationally tractable, economically interpretable, and robust in the paper's appendix to reversing the update order.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies demand

Demand is identified from three ingredients.

- **Cross-sectional product differentiation.** Different trucks have different GWR, cab design, and long-option configurations.
- **Price variation from wages and competition.** Wages shift costs; local product overlap shifts equilibrium markups.
- **Micro moments.** Buyer-type-by-product-type matching pins down which buyers value which truck characteristics.

This is especially important because the paper wants economically meaningful substitution patterns, not just fit to market shares.

4.2 What identifies sunk costs

The most difficult part of the paper is stage-1 identification.

- **Construction demand shifter.** When construction-related potential buyers expand, products in the medium GWR range become much more profitable. The paper uses these states to identify product entry and continuation in those segments.
- **Cab-length deregulation.** When regulations relaxed length constraints, the relative attractiveness of heavy cabover trucks declined. The paper uses these states to identify exit from heavy cabover segments and entry into substitute conventional segments.

Why this is clever. Entry and exit are otherwise too rare and too selected to be directly informative. The paper uses settings where the data almost force certain product changes to occur somewhere, while the shocks themselves are plausibly exogenous to the sunk-cost disturbances.

4.3 Why the reduced-form evidence matters

Figures 2 and 3 are not just descriptive. They are effectively the first-stage evidence for the sunk-cost estimation strategy.

- Figure 2 shows that more construction buyers coincide with more offerings in the medium-duty GWR range.
- Figure 3 shows that cab-length deregulation coincides with the decline and eventual disappearance of heavy cabover offerings.

If firms did *not* adjust product menus to these shifts, the structural model would be much less believable.

4.4 Why the second-stage estimates look credible

The paper checks whether the demand and supply estimates produce sensible economic objects.

- All own-price elasticities are above 1, as they should be under profit-maximizing pricing.
- For low-GWR vehicles in 1992, average own-price elasticities are around **10.9**, somewhat higher but broadly in line with the passenger-car literature.
- Implied operating margins are around **6.1%–6.6%**, close to reported average operating margins of **7.5%**.

That is not proof the model is right, but it is a strong sanity check.

4.5 What makes the quantitative exercise credible

- The market is unusually well suited for product-level entry/exit analysis because product changes are frequent and technologically feasible.
- Buyer heterogeneity is directly observable in microdata, which gives the demand system real economic content.
- The policy experiment is concrete: the 2009 GM/Chrysler rescue and realistic acquisition/liquidation alternatives.
- The author separates price-only counterfactuals from price-plus-repositioning counterfactuals, which makes the economic mechanism transparent.

4.6 Main limitations

- **Transaction prices are not observed.** The paper uses MSRP.
- **The analysis is only for the commercial segment.** It cannot by itself settle the full \$85 billion auto bailout question.
- **Stage-1 estimates are set identified, not point identified.**
- **Counterfactual product equilibria are selected via an iterative learning process, not by exhaustive equilibrium enumeration.**
- **Employment implications are suggestive.** They rely on approximate linearity between vehicles produced and workers employed.

These are genuine limitations, but they do not undercut the main message that endogenous product responses matter a lot.

4.7 Relation to the literature

The paper sits at the intersection of several literatures.

- **BLP-style demand estimation:** differentiated-product demand with rich heterogeneity.
- **Entry/exit and positioning:** Seim, Ho, Fan, Eizenberg, and related work.
- **Dynamic IO:** Markov-perfect and product-positioning games with large choice sets.
- **Policy analysis:** merger, bankruptcy, and bailout counterfactuals.

Its distinctive contribution is to show that *model-level* entry and exit can overturn price-only policy predictions even when *firm-level* entry is absent.

5 Tables and figures

5.1 Figure 1 and Table 1: what is special about this market

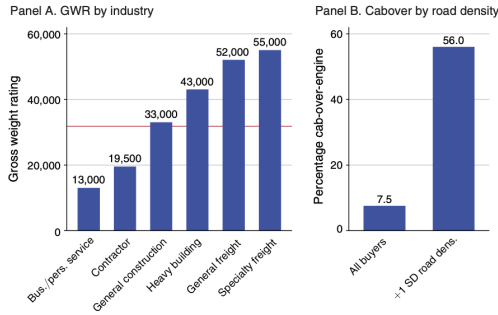


FIGURE 1. AVERAGE PRODUCT CHARACTERISTICS CONDITIONAL ON PURCHASE AND BUYER TYPE

Notes: This figure illustrates demand heterogeneity. Panel A shows expected GWR conditional on purchase across the main US Census-defined industries. Panel B shows cab-over-engine (cabover) sales as a proportion of all sales either overall or for buyers whose USDOT road density measure is one standard deviation above the mean for that measure.

TABLE 1—SUMMARY STATISTICS					
	Min.	Max.	Average over each period		
			1986–1994	1995–2003	2004–2012
<i>Panel A. Count of product offerings by type</i>					
Light-medium GWR conventional	20	24	21.0	21.7	22.3
Medium GWR conventional	14	27	25.0	20.8	20.4
Heavy GWR conventional	28	36	31.6	33.0	32.9
Light-medium GWR cabover	8	11	10.0	10.3	9.8
Medium GWR cabover	2	12	11.0	8.1	5.1
Heavy GWR cabover	0	8	7.4	1.8	0.0
Compact-front-end (all GWRs)	3	5	3.0	3.3	4.1
Long/extended cab (all GWRs)	9	16	15.4	11.6	9.7
All types	62	85	77.6	75.4	75.7
	Min.	Max.	Average over each period		
			1992–1994	1995–2003	2004–2012
<i>Panel B. Prices charged and units sold</i>					
Price (in 2005 \$)	\$70,324	\$75,588	\$73,075	\$73,448	\$72,104
Quantity (in 000s of units)	165,678	584,057	286,246	395,879	369,985
	Min.	Max.	Average over each period		
			1986–1994	1995–2003	2004–2012
<i>Panel C. Proportion of potential vehicle buyers by type (%)</i>					
Freight-related industries	22.1	32.1	25.2	28.0	28.6
Construction-related industries	36.5	50.4	43.3	45.6	44.7
Business- and personal service-related industries	22.8	37.1	31.5	26.4	26.8

Notes: The first panel counts the number of offerings by year in each category. For example, there were a minimum of 20 vehicles offered in the “light-medium GWR conventional” category. The second panel summarizes average price and total units sold by year, and the third panel summarizes the average buyer attribute by year. For price and quantity variables, the third column is truncated due to data restrictions, i.e., that price and quantity in the panel begin in 1992 (see Section I for more details). The low average quantity for the first period, 1992 to 1994, reflects the recession of the early 1990s. Aggregate sales data from *Ward’s Automotive Yearbook* confirm that average unit sales for the (extended) period 1986–1994 would be in line with the periods 1995–2003 and 2004–2012.

Read: Figure 1 shows the mapping from buyer type to truck characteristics; Table 1 shows that the menu of products moves a lot over time.

- Figure 1, panel A: expected GWR is low for service-type buyers, middle for construction, and high for freight.
- Figure 1, panel B: cabovers are rare overall but much more important in dense, urban settings.
- Table 1 shows that offerings vary from **62** to **85** total models across the sample.
- The market is therefore not one in which firms sit on fixed menus for decades.

Why it matters. This is the core empirical fact that justifies the whole paper. If product menus were rigid, a price-only merger model might be fine. Here they are clearly not rigid.

5.2 Figure 2 and Figure 3: reduced-form evidence for product repositioning

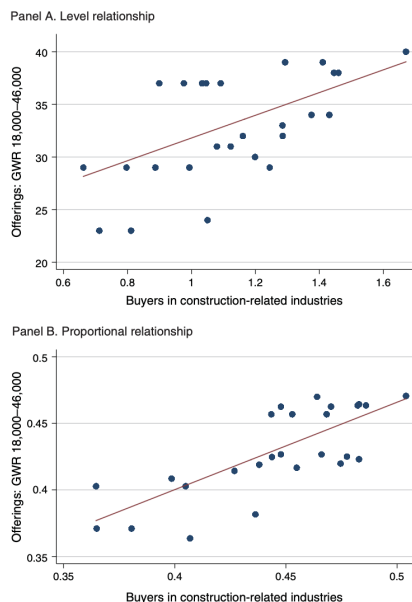


FIGURE 2. REDUCED-FORM EVIDENCE FROM THE CONSTRUCTION INDUSTRY

Notes: The figure reports the link between the changing composition of potential buyers and changing product offerings. The x-axis measures the total number of potential buyers in millions while the y-axis counts product offerings (in panel A) and measures the proportion of total product offerings (panel B). The subset of products with GWR in [18,000,46,000] is chosen because this is where demand is concentrated for construction-related buyers.

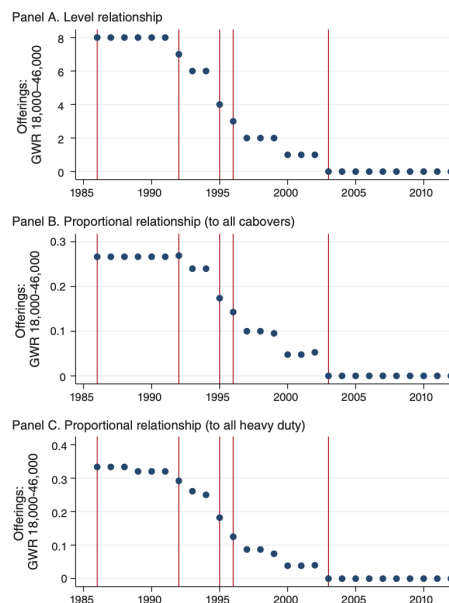


FIGURE 3. REDUCED-FORM EVIDENCE FROM THE FREIGHT INDUSTRY

Notes: This figure reports the link between regulatory decisions and changing product offerings. On the x-axis is the year, and vertical lines represent regulatory events that relax the cab length laws. Each y-axis measures the number of heavy cabover vehicle offerings. In panel A, this measure is in levels. In panels B and C, this measure is as a proportion of all cabover vehicles and all heavy vehicles, respectively.

Read: These figures are the first-stage evidence that firms adapt product offerings to market-specific shocks.

- Figure 2: as the number or share of construction-related buyers rises, medium-duty offerings rise too.
- Figure 3: as cab-length regulation is relaxed, heavy cabover offerings shrink and eventually become obsolete in the United States.

Interpretation.

- Construction demand identifies entry and continuation in medium GWR products.
- Cab-length deregulation identifies exit from heavy cabovers and entry toward substitute products.

Why it matters. These are the exogenous demand shifters that let the author say something disciplined about sunk costs.

5.3 Table 2: second-stage parameter estimates

Read: Table 2 gives the demand and marginal-cost parameters.

- The average buyer likes higher GWR: coefficient **2.298**.

- The average buyer strongly dislikes cabovers: coefficient about **-24.805**.
- Price enters negatively: coefficient about **-0.44**.
- Freight and construction buyers value GWR much more than average buyers.
- Urban buyers strongly prefer cabovers: cabover $\times$ urban coefficient about **35.053**.

On the cost side:

- higher GWR raises marginal cost,
- compact-front-end lowers marginal cost slightly relative to conventional,
- long option raises cost,
- wages matter positively.

Why it matters. The interaction estimates line up tightly with industry common sense and the microdata. That gives confidence that the substitution patterns used in the counterfactuals are not mechanical artifacts.

TABLE 2—SECOND-STAGE PARAMETER ESTIMATES

	Parameter estimate	Standard error
Demand		
Mean parameters		
GWR (in 0,000s of lbs.)	2.298	0.873
Indicator for cabover	-24.805	2.026
Indicator for compact-front-end	-1.425	0.273
Indicator for long option	1.026	0.424
Price (in 000s of 2005\$)	-0.44	0.037
Constant	0.066	0.32
Buyer attribute interactions		
GWR $\times$ specialty freight industry	23.023	1.574
GWR $\times$ general freight industry	16.434	0.553
GWR $\times$ heavy building industry	11.162	0.495
GWR $\times$ general construction industry	7.35	0.896
GWR $\times$ contractor industry	3.693	0.915
Indicator for cabover $\times$ general freight industry	0.906	0.261
Indicator for cabover $\times$ general freight industry $\times$ length laws	-1.183	0.881
Indicator for cabover $\times$ urban	35.053	2.283
Indicator for compact-front-end $\times$ urban	0.945	0.725
Indicator for long option $\times$ freight industry $\times$ z_{lr}^*	5.842	1.754
Constant $\times$ urban	-5.67	1.13
Constant $\times$ specialty freight industry	-92.802	3.578
Constant $\times$ general freight industry	-57.517	2.233
Constant $\times$ heavy building industry	-29.843	1.557
Constant $\times$ general construction industry	-15.008	0.988
Constant $\times$ contractor industry	-6.993	0.87
Constant $\times$ z_{lr}^*	6.288	0.537
Marginal costs		
GWR	0.385	0.06
Indicator for cabover	0.002	0.033
Indicator for compact-front-end	-0.089	0.04
Indicator for long option	0.073	0.012
Wage (\$ per hour)	0.016	0.004
t (year count)	0.041	0.053
Constant	2.132	0.215
Observations	1,575	

Notes: Specialty freight, general freight, heavy building, general construction, and contractor industry denote indicators for the industry in which the buyer operates. Urban denotes the road density measure. Length laws denote the current cumulative number of legislative and court decisions related to regulation of cab length. z_{lr}^* and z_{lr}^* are drawn from standard normal distributions, so their coefficients correspond to the standard deviation of their distributions. The marginal cost coefficients for the wage and the year count differ from the prior draft due to inadvertent rescaling in that draft.

TABLE 3—SUNK COST PARAMETER ESTIMATES

	95 percent confidence interval
Sunk cost parameters (in 000s of 2005\$)	
Mean parameter	
GWR (in 0,000s of lbs.)	[\$3,163, \$4,664]
Indicator for cabover	[5,052, 7,478]
Indicator for compact-front-end	[-8,852, 3,411]
Indicator for long option	[-6,339, -3,433]
Constant	[-27,145, -21,103]
Firm identity interactions	
Constant $\times$ indicator for <i>Big3</i>	[-3,066, 7,732]
GWR $\times$ indicator for <i>Big3</i>	[-2,143, 299]
Constant $\times$ indicator for <i>Jpn</i>	[-5,204, -400]
Indicator for cabover $\times$ indicator for <i>Jpn</i>	[9, 5,289]
Scrap value scale parameter ($-\lambda$)	[0.531, 0.613]
Observations	11,123
Moments	97

Notes: Sunk costs are set identified by moment inequalities, so bounds are reported. Confidence intervals contain the true parameters with probability 0.95. The final parameter, equal to $-\lambda$, is the proportion of the sunk costs of introducing a model that the firm recaptures when it retires the model, i.e., it “scales” sunk costs to scrap values. “*Big3*” denotes indicator for Ford, General Motors, or Chrysler brand. “*Jpn*” denotes indicator for Isuzu, Hino, Bering, Mitsubishi Fuso, or UD/Nissan Diesel brand. Observations reflect the number of unique comparisons between first-stage and second-stage payoffs on which the moments are based.

5.4 Table 3: sunk-cost estimates

Read: Table 3 reports set-identified bounds for product entry costs and scrap values.

- For a conventional truck at zero GWR, entry costs are roughly **\$5 million to \$25 million** using the midpoint interpretation described in the text.
- The compact-front-end is more expensive to introduce than a conventional truck.

- The long-cab option is also more expensive.
- The cabover is relatively inexpensive to introduce.
- Firms recover about **55%–60%** of sunk costs when they retire a model.

There is also systematic firm heterogeneity:

- Japanese firms have an advantage in introducing cabovers.
- Big Three firms have an advantage in lower-GWR vehicles, plausibly because of spillovers from their passenger-vehicle operations.

Why it matters. The key policy result only works if these product introduction costs are *low enough* that rivals really will respond to local increases in market power. Table 3 shows that the data are consistent with exactly that.

5.5 Table 4 and Figure 4: Ford acquisition as the canonical example

TABLE 4—GENERAL MOTORS AND CHRYSLER PRODUCTS IN 2009

Parent	Brand	GWR	Conv.	Cabover	Compact-front-end	Long/extended
Chrysler	Dodge	12,000	×			
Chrysler	Dodge	13,500	×			
Chrysler	Dodge	16,000	×			
Chrysler	Dodge	19,500	×			
GM	Chev./GMC	12,000			×	
GM	Chev./GMC	12,000	×			
GM	Chev./GMC	13,500	×			
GM	Chev./GMC	14,500		×		
GM	Chev./GMC	16,000	×			
GM	Chev./GMC	19,500		×		
GM	Chev./GMC	19,500	×			
GM	Chev./GMC	21,500	×			
GM	Chev./GMC	28,000	×			
GM	Chev./GMC	33,000		×		
GM	Chev./GMC	40,000	×			

Notes: General Motors and Chrysler together offer 15 products approximately at the point the bailout decision is made. As in the rest of the paper, GMC and Chevrolet are treated as one brand, since they are physically identical (except for the brand badge) and indistinguishable in the product characteristic guides and pricing data.

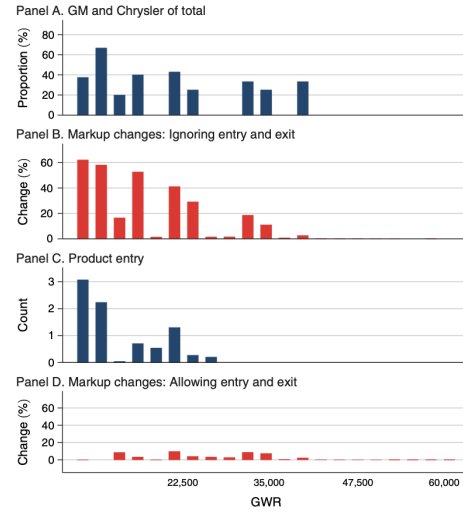


FIGURE 4. COUNTERFACTUAL OUTCOMES UNDER ACQUISITION BY FORD

Notes: The x-axis is the GWR (likely the single most important characteristic). The y-axis in panel A is the proportion of all products that are offered by GM and Chrysler. The y-axis in panel B is percentage change in dollar markup, i.e., $(P_{FordAcquisition} - mc) / (P_{Bailout} - mc)$, for the most affected product in each GWR bin (if the economic model does not allow for product repositioning). The y-axis in panel C is a combined count of the net number of product entries. The y-axis in panel D is analogous to the second (but the economic model allows for product repositioning).

Read: Table 4 lists the **15** GM and Chrysler products active around the 2009 bailout decision. Figure 4 then shows what happens if Ford acquires them.

- Table 4 makes clear that GM and Chrysler are concentrated in low- and medium-GWR segments.
- Figure 4, panel A shows where GM and Chrysler's product presence is strongest.
- Figure 4, panel B shows that, if product offerings are held fixed, markups spike exactly in those GWR bins.
- Figure 4, panel C shows that surviving rivals enter *precisely* where those markup increases are largest: about **7–8** new products on average.

TABLE 5—COUNTERFACTUAL OUTCOMES

	Product entry and exit ignored			Product entry and exit allowed		
	Ford acq.	PACCAR acq.	Liq.	Ford acq.	PACCAR acq.	Liq.
Markups (percent)						
Most affected model	62.1	23.0	26.9	16.8	15.4	6.2
Most affected vehicle type	53.2	13.0	32.3	9.6	7.6	5.3
Market	10.9	3.0	4.0	0.8	0.5	-0.0
Output (percent)						
Most affected model	-34.4	-18.0	NR	6.7	16.5	8.9
Most affected vehicle type	-30.0	-8.6	-100.0	-63.9	-54.9	-64.7
Market	-5.1	-1.4	-11.2	-1.3	-1.3	-1.6
Compensating variation	119.0	33.0	253.0	22.0	26.0	28.0

Notes: This table compares market outcomes across counterfactual policy choices. The left three columns report predictions from an economic model that ignores product entry and exit. The right three columns report predictions from one that allows for this behavior. The first three rows report markup changes for the most affected model, the most affected product type (averaged over models of the same type), and the market overall (averaged across models). The next three rows are identical to the first three, except that they report output changes rather than markup changes. The final row reports compensating variation (for the counterfactual policy relative to the bailout). NR denotes not relevant. Compensating variation is expressed in millions of 2005\$.

- Figure 4, panel D shows the result: once product entry and exit are allowed, markup increases collapse.

Why it matters. This figure is the entire paper in one picture. The place where policy appears most harmful under a fixed-menu model is the place where endogenous entry is most aggressive.

5.6 Table 5: the full policy comparison

Read: Table 5 is the headline quantitative result.

Ignoring product entry and exit.

- **Ford acquisition:** most affected model markup + **62.1%**; market output – **5.1%**; compensating variation **\$119 million**.
- **PACCAR acquisition:** most affected model markup + **23.0%**; market output – **1.4%**; compensating variation **\$33 million**.
- **Liquidation:** most affected model markup + **26.9%**; market output – **11.2%**; compensating variation **\$253 million**.

Allowing product entry and exit.

- **Ford acquisition:** most affected model markup + **16.8%**; market output – **1.3%**; compensating variation **\$22 million**.
- **PACCAR acquisition:** most affected model markup + **15.4%**; market output – **1.3%**; compensating variation **\$26 million**.
- **Liquidation:** most affected model markup + **6.2%**; market output – **1.6%**; compensating variation **\$28 million**.

Important reading note. In the entry/exit columns, output for the “most affected model” can be positive. This is not a contradiction. It means some surviving or newly entered models gain demand even while total market output still declines slightly.

Why it matters. Table 5 is the sharpest statement of the paper’s message: price-only counterfactuals massively overstate the effect of acquisition or liquidation.

5.7 Policy significance beyond prices

The paper also translates the output effects into broader policy terms.

- If employment is roughly proportional to output, ignoring entry and exit overstates lost wages by more than **\$480 million** annually.
- The commercial-vehicle share of the rescue is about **\$6 billion**.
- Under liquidation, the break-even discount rate is about **4.7%** if entry/exit is ignored but only **0.5%** if it is allowed.
- If the planner also cares about persistent wage losses, the range in which accounting for entry/exit is pivotal widens to about **2.0%–15.3%**.

Interpretation. The bailout decision itself can change once product repositioning is properly taken into account.

6 Short summary

- The paper builds a structural model where firms choose *products first, prices second*.
- Buyer heterogeneity is central: different industries and geographies want different truck characteristics.
- Product entry and exit are identified using construction-demand shifts and truck-length deregulation.
- Estimated sunk costs are low enough that rivals respond strongly to local increases in profitability.
- Because of that, fixed-product counterfactuals dramatically overstate the effects of acquisition or liquidation on markups, output, and welfare.
- The commercial-vehicle segment evidence implies that endogenous product repositioning can be decisive for policy evaluation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that in differentiated product industries with *stable firms but flexible product menus*, policy counterfactuals are badly misspecified if they hold product characteristics fixed. In the commercial vehicle market, the effect is first order: product entry and exit offset markup increases and output losses by up to roughly three-quarters.

7.2 Economic intuition

The intuition is simple and powerful. Policies like merger, acquisition, or liquidation do two things at once:

1. they increase market power where the departing or merged firms had strong product overlap;
2. they create exactly the profit opportunities that surviving rivals need in order to introduce new products.

If sunk costs are not too large, the second force disciplines the first. More product variety makes substitution easier, reducing the post-policy markup spike.

7.3 Takeaway

This is a paper about *where equilibrium lives*. In many industrial-organization applications, the relevant equilibrium object is not just prices conditional on a fixed product set. It is prices *and* product offerings together. Wollmann shows that once product menus are allowed to move, some seemingly dramatic policy effects become much smaller—and sometimes small enough to flip the policy conclusion itself.